AMR as a promising automatic evaluation metric for machine translation: Part II

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Abstract

Abstract Meaning Representation (AMR) represents the meaning of a sentence as a rooted directed graph and has been used to support many meaning-focused NLP tasks. This study extends my previous work on AMR-based metrics for machine translation (MT) quality evaluation, tested on the WMT23 Metrics Shared Task, by adding three AMR similarity metrics from different categories, *SemBleu*, *Smatch++*, and *AMRsim*, and a considerably stronger AMR parser, *StructuredBART-MBSE*. A better parser and a better similarity metric together improve the performance of AMR-based metrics, raising their rank in the WMT23 evaluation from 28th to 23rd place. AMR graph-based metrics outperform all text-based metrics that rely on lexical overlap, as well as several embedding-based metrics and a fine-tuned metric. However, they remain behind the embedding-based MT metric baseline *BERTscore* and the top fine-tuned metrics. The data and implementation details are available in the GitHub repository.

1 Introduction

Abstract Meaning Representation (AMR) is a semantic framework that represents the concepts in a sentence and the semantic relations between them as a rooted,directed and acyclic graph, a structure that is both human-readable and machine-processable. It has proved useful in many natural language processing (NLP) applications, especially meaning-focused tasks such as information retrieval (Rao et al., 2017), question answering (Kapanipathi et al., 2021), paraphrase detection (Issa et al., 2018; Sousa and Cardoso, 2025; Park et al., 2025), summarization (Hardy and Vlachos, 2018; Hua et al., 2022), and machine translation (Linfeng et al., 2019; Nguyen et al., 2021; Li and Flanigan, 2022; Zhou and Zhang, 2026).

These applications use the AMR graphs of input sentences as additional semantic information to improve task performance. In contrast, the author's previous work used AMR as an evaluation tool to assess the quality of model outputs, specifically the quality of machine translation (MT), using the data and comparison methods of the WMT23 Metrics Shared Task (Freitag et al., 2023). The results show that even with only the most basic standard AMR similarity metric, *Smatch* (Cai and Knight, 2013), and a moderately performing AMR parser, the AMR graph-based metric already outperforms all text-only, string-based MT metric baselines and some embedding-based ones (see Table 6 and Figure 1).

However, the performance of AMR-based metrics depends heavily on the quality of the graphs and on the similarity measure used. A better AMR parser produces more accurate graphs from sentences, and a better AMR similarity metric captures more of the semantic similarity between graph pairs. Both contribute to a more reliable evaluation of translation quality and to assessments that align better with human judgments. Therefore, this study extends the previous one by adding three more AMR similarity metrics and a considerably stronger AMR parser, in order to examine more comprehensively how well AMR graph-based metrics perform in MT quality evaluation.

The three additional AMR similarity metrics, *SemBleu*, *Smatch++*, and *AMRsim*, come from different categories and are introduced in Section 1.2. The second AMR parser, *StructuredBART-MBSE*, is introduced in Section 1.1. Section 2 describes the implementation details of these new metrics and the new

parser. Section 3 presents and discusses the results, and Section 4 concludes the paper.

1.1 AMR Parsers Overview

The development of AMR parsers can be broadly divided into three stages: early alignment- and transition-based parsers, fine-tuned transformer-based parsers, and, more recently, parsers that prompt large language models (LLMs).

Alignment-based AMR parsers construct AMR graphs by learning alignments that map spans in the input text to the corresponding concept nodes and relations within its AMR graph, often with help of rule-based heuristics. A representative system is JAMR (Flanigan et al., 2014), which uses rule-supported a semi-Markov model to align spans in the input text to graph nodes and identifies relations by selecting the highest-scoring connected subgraph.

Transition-based parsers construct AMR graphs incrementally by applying a learned sequence of parse transitions. These transitions are induced from training data: the sentence is first converted into a dependency tree, and the parser then learns a transition system (a set of transformation actions) that maps the dependency structure to an AMR graph. Representative systems include AMR-EAGER (Damonte et al., 2017) and CAMR (Wang et al., 2016). Alignment- and transition-based parsers typically achieve *Smatch* F1 scores in the range of 58–67 on the AMR corpus LDC2015E86 test split (Damonte et al., 2017), which are later substantially surpassed by transformer-based parsers.

With the rise of neural network methods, researchers also explored neural architectures for AMR parsing. These approaches typically cast AMR parsing as a sequence-to-sequence task using an encoder–decoder architecture: the input is the sentence, and the output is a linearized AMR graph, most often produced via depth-first traversal. Various preprocessing techniques were introduced to mitigate data sparsity, and postprocessing steps were required to reconstruct full AMR graphs from the generated linearized sequences.

Early neural attempts, however, achieved limited success, likely due to the small size of available sentence–AMR training pairs. These models relied on word- or character-level bidirectional recurrent neural networks (RNNs) (Barzdins and Gosko, 2016; Peng et al., 2017) or stacked RNN encoders (Konstas et al., 2017), paired with standard attention-based decoders. Their performance remained below that of alignment-based parsers such as JAMR and transition-based parsers such as CAMR on the AMR corpus LDC2015E86 (Konstas et al., 2017).

With the introduction of pre-trained Transformer-based models, AMR parsing performance improved substantially, because these models already encode extensive linguistic knowledge learned from large corpora, reducing the need for AMR-specific training data and helping overcome the task's severe data-sparsity limitations. Although the earliest Transformer approaches are not strictly sequence-to-sequence parsers since they directly predict graph nodes, edges, and alignments rather than linearized AMR sequences, they are included here because they are among the first ones to incorporate pre-trained BERT (Devlin et al., 2019) representations, which led to significant gains over earlier neural baselines. Models such as (Zhang et al., 2019) and (Cai and Lam, 2020) surpassed 70 Smatch on AMR 1.0 corpus (LDC2014T12) and exceeded 80 on AMR 2.0 (LDC2017T10).

The adoption of pre-trained BART (Lewis et al., 2019) pushes performance even further: instead of using only its encoder to represent input sentences, BART-based parsers apply full transfer learning by fine-tuning the pre-trained encoder–decoder model on labeled AMR data. BART—**B**idirectional and **A**uto-**R**egressive **T**ransformer—is a denoising autoencoder that corrupts an input and reconstructs it using a BERT-like encoder and a GPT-like decoder. This architecture is particularly well suited for AMR parsing, as the input sentence and its AMR graph share substantial structural and semantic information. Current AMR parser baselines such as *SPRING* (Bevilacqua et al., 2021) and *AMR-BART* (Bai et al., 2022) both rely on transfer learning from BART. The key difference is that *SPRING-silver* is initialized from a pre-trained BART model using only silver graph data generated by *SPRING* as input.

AMRBART, instead, jointly pretrains BART on textual data and large amounts of silver AMR graphs produced by *SPRING*, using both node/edge and subgraph denoising objectives. These graph-aware pre-trained BART model is then fine-tuned on the AMR 2.0 and AMR 3.0 (LDC2020T02) training sets. Their performances based on BART Large configuration, reach 84.3 and 85.4 on AMR 2.0 test set, and 83.0 and 84.2 on AMR 3.0 test set (Bai et al., 2022).

Other attempts to improve AMR parsing with BART continued to incorporate graph information during training. *StructuredBART* (Zhou et al., 2021b) follows transition-based AMR parsing and fine-tunes the BART Large model to predict the next parsing action rather than a linearized graph. It repurposes one decoder self-attention head as a pointer to select which earlier node to connect, and is trained on sentence-action pairs with an added pointer loss to supervise this mechanism. On AMR 2.0, it achieves 84.3 *Smatch* using text-only pre-trained BART Large and 84.7 when initialized from silver-graph-pretrained BART Large, marginally surpassing the non-transition-based *SPRING-silver*. However, it still falls below *AMRBART*, which benefits from unified text-and-graph pre-training.

(Lee et al., 2022) recognize the importance of data augmentation and aim to improve the quality of silver AMR graphs used in pre-training. They achieve this through an ensemble-distillation method that selects the best graph among multiple parser outputs, using their Maximum Bayes Smatch Ensemble (MBSE) technique. The resulting high-quality silver graphs are then used to pre-train *StructuredBART*. With silver data ensemble-distilled from five AMR parsers: *APT* (Action-Pointer Transformer) (Zhou et al., 2021a), *StructuredBART*, two *SPRING* from different random seeds, and *AMRBART*. Their model achieves *Smatch* scores up to 85.9 on AMR 2.0 and 84.3 on AMR 3.0, outperforming all previous parsers.

LeakDistill (Vasylenko et al., 2023) is another powerful AMR parser that integrates graph information during training. It also builds on pre-trained BART and incorporates structural information through self-knowledge distillation. During training, graph structure is "leaked" into the encoder by inserting a graph adapter after each layer, which encodes the gold AMR graph together with word-node alignments. The decoder receives both the sentence-only encoder representations and the graph-leaked encoder representations, and a Kullback–Leibler divergence loss is applied to "distill" these leaked structural signals when generating the linearized AMR graph. *LeakDistill*, built on BART Base, already achieves performance comparable to *SPRING* and *AMRBART* on both AMR 2.0 and AMR 3.0. When scaled to BART Large and pre-trained with silver data, it surpasses all non-ensemble AMR parsers, reaching 86.1 on AMR 2.0 and 84.6 on AMR 3.0.

The most recent development in AMR parsing leverages large pre-trained language models (LLMs). (Lee et al., 2023) fine-tune sentence and linearized graph pairs on the T5 (Text-to-Text Transfer Transformer) model, specifically the instruction-tuned Flan-T5 variant (Chung et al., 2022). T5 models are also encoder–decoder architectures like BART, but trained on different objectives and data. The instruction-fine-tuned Flan-T5 prepends task-specific prefixes such as "summarize" to inputs. Using Flan-T5 Large additionally pre-trained on silver AMR data, (Lee et al., 2023) report *Smatch* scores of 85.8 on AMR 2.0 and 84.6 on AMR 3.0, approaching the performance of *StructBART-MBSE*. With further hyperparameter tuning, their model reaches 86.1 on AMR 2.0 and 84.7 on AMR 3.0.

(Ho, 2025) fine-tune four open-source decoder-only LLMs and show that generative models trained on AMR graph data can effectively perform AMR parsing, achieving *Smatch* scores between 0.78 and 0.80 on AMR 3.0, approximating the performance of *SPRING* without silver-data pre-training.

In contrast, work that evaluates decoder-only LLMs under zero-shot or few-shot prompting consistently shows that models without any graph-specific fine-tuning perform poorly on AMR parsing. Under zero-shot settings, performance remains particularly weak, while some models achieve *Smatch* scores between 0.7 and 0.8 only when provided with instructional prompts and a small number

of demonstrations (e.g., five-shot prompting) (Li and Fowlie, 2025; Buljan et al., 2026). These results suggest that current decoder-only LLMs are not yet viable candidates for standalone AMR parsing.

1.2 AMR Similarity Metrics Overview

Since *Smatch* (2013), new AMR similarity metrics have continued to emerge to address its limitations and to better capture semantic similarity between AMR graphs. Although *Smatch* remains the canonical AMR similarity metric, it is not without issues. Its main limitations are:

(1) It treats an AMR graph as an unordered collection of nodes and edges, performing one-to-one mappings without accounting for the relational structure or dependencies among them.

(2) It determines the triplet alignment by searching over possible variable assignments for node concepts and selecting the one that yields the highest F1 score. This search procedure is computationally inefficient, lacks reproducibility, and—crucially—offers no guarantee of finding the optimal alignment.

(3) As a node-to-node and edge-to-edge string-matching metric, it cannot capture fine-grained semantic similarity beyond surface symbols. For instance, mapped nodes whose concepts are synonyms are semantically closer than non-synonyms, but *Smatch* treats them as equally distinct.

Due to limitation (1), two semantically distant sentences, with different concepts (node content) but similar edge labels, or with similar concepts but different edge labels, can receive high *Smatch* scores (Anchieta et al., 2019; Song and Gildea, 2019). Limitation (2) further introduces alignment errors and assigns inconsistent weights to different types of mismatches (Anchieta et al., 2019).

SEMA (Semantic Evaluation Metric for AMR) (Anchieta et al., 2019) and *SemBleu* (Song and Gildea, 2019) are two AMR similarity metrics designed to address the first two limitations. *SEMA* employs a Breadth-First Search (BFS) strategy to traverse both AMRs and directly compare their node concepts, without requiring variable alignment, while also evaluating the relations between nodes (i.e., edges) in traversal order to capture dependencies among neighboring nodes. *SemBleu* likewise uses BFS, but extends the distance of dependency (longer than two node sequences) by employing n-grams, where n corresponds to the number of nodes connected in the graph. For example, each pair of directly connected nodes together with their relation (edge label) constitutes a bi-gram, for example *ask-01 :ARG0 girl*. Both metrics capture semantic similarity between AMR graphs more effectively by incorporating node–edge dependencies, and they are more efficient because no search over node mappings is required. *SemBleu* is additionally reported to correlate more strongly with human evaluations (Song and Gildea, 2019).

However, *SemBleu* does not come without problems. (Opitz et al., 2020) establish seven principles for AMR graph similarity comparison and argue that *SemBleu* is less adequate than *Smatch*, as it severely violates four of these principles: *identity of indiscernibles*, *symmetry*, *no bias*, and *graded semantic match*, whereas *Smatch* practically violates only the last one (with mild or no violation of the others). *SemBleu* can assign a maximum score (1.0), indicating total equivalence, to two AMR graphs that are not equivalent, e.t., *The man see the other man* vs. *The man see himself*. It is also asymmetric as a metric, such that *SemBleu(A, B)* != *SemBleu(B, A)*. Moreover, due to the n-grams extracted from the BFS traversal, it penalizes mismatches involving high-degree nodes (i.e., nodes with many outgoing or incoming edges) more harshly than mismatches involving low-degree nodes, since high-degree nodes appear more frequently in the extracted n-grams. For example, *SemBleu* considers *In April, a woman travels a car from Rome to Pisa* more similar to *In April, a sailor travels with a ship from Porto to Natal* than to *In April, a woman drives a car from Rome to Pisa* (Opitz et al., 2020). Most importantly, like *Smatch*, it does not incorporate fine-grained semantic similarity beyond string (symbol) matching.

Parser	Type	Silver	A1	A2	A3
JAMR (Flanigan et al., 2014)	Align.	N	58.0*	–	–
CAMR (Wang et al., 2016)	Trans.	N	66.5*	–	–
AMR-EAGER (Damonte et al., 2017)	Trans.	N	64*	–	–
CHAR-LSTM (Barzdins and Gosko, 2016)	RNN	N	43*	–	–
LSTM (Peng et al., 2017)	RNN	N	52*	–	–
Stacked-LSTM (Konstas et al., 2017)	RNN	N	62.1*	–	–
seq2graph (Zhang et al., 2019)	BERT	N	71.3	77	–
seq2graph (Cai and Lam, 2020)	BERT	N	75.4	80.2	–
SPRING-silver (Bevilacqua et al., 2021)	BART-Large	Y	–	84.3	83.0
AMRBART (Bai et al., 2022)	BART-Large	Y	–	85.4	84.2
APT (Zhou et al., 2021a)	action-pointer+RoBERTa-Large	Y	–	82.6	83
StructuredBART (Zhou et al., 2021b)	action-pointer+BART-Large	Y	–	84.7	82.6
StructuredBART-MBSE (Lee et al., 2022)	action-pointer+BART-Large	Y	–	85.9	84.3
LeakDistill (Vasylenko et al., 2023)	BART-Large	Y	–	86.1	84.6
Flan-T5 Large, tuned (Lee et al., 2023)	T5	Y	–	85.8	84.6
Fine-tuned LLMs (Ho, 2025)	LLM	N	–	–	78–80
Prompted LLMs (Li and Fowlie, 2025) (Buljan et al., 2026)	LLM	N	0-shot:14–60, 5-shot:60–80		

Table 1: Overview of the AMR parsers discussed, with *Smatch* F1 on the AMR 1.0 (A1), 2.0 (A2), and 3.0 (A3) test sets. Note that results obtained using silver data are not directly comparable, as the silver pre-training datasets differ in size. Align.: alignment-based; Trans.: transition-based; "N": No; "Y": Yes; "–": not reported; * marked: based on AMR test set of LDC2015E86 corpus.

They therefore introduce S^2*match* (Soft Semantic Match) to fully satisfy all seven principles. It builds on *Smatch* but additionally incorporates semantic similarity between concepts in instance triplets (soft matching) by using their GloVe (Pennington et al., 2014) word vectors. This extension includes the graded semantics and increases the capacity of the metric to assess AMR graphs from different sentences, which is crucial when AMR is applied beyond parser evaluation—where AMRs typically represent the same sentence—such as in cross-lingual AMR parsing and paraphrase detection or generation. S^2*match* is shown to align slightly better with human judgement of AMR graph similarity, although the alignment remains far from perfect (Opitz et al., 2020). (Wein and Schneider, 2022) extended it for cross-lingual AMR evaluations as *XS*^{2}*match*.

However, S^2*match* still does not solve the core *Smatch* limitation of excluding semantic similarities between substructures (i.e., structural dependency). Opitz et al. recognize this issue and adapt *WWLK* (*Wasserstein Weisfeiler–Leman Kernel*) (Togninalli et al., 2019) to AMR graph similarity the following year. They first propagate node embeddings iteratively to obtain contextualized representations of the graph, and then employ *WWLK* to compute the minimum-cost distance for transforming one graph into another. In this way, the metric successfully incorporates graded similarities between graph substructures without the high-degree-node bias observed in S^2*match* (Opitz et al., 2021). This unifies the strengths of BFS-based metrics (*SEMA*, *SemBleu*) and alignment-based metrics (the *Smatch* variants) while mitigating their respective weaknesses. Its variant $WWLK_\theta$ introduces an edge parameter θ to further account for the influence of semantic relations (e.g., *negation*) in the final graph-embedding representation. Most importantly, the *WWLK* variants achieve substantially stronger correlation (over 10% improvement) with human similarity ratings than all previous metrics, as demonstrated on the AMR benchmark evaluation datasets STSB (Baudiš et al., 2016) and SICK (Marelli et al., 2014) (Opitz et al., 2021).

The refinement of *Smatch* continued. (Opitz, 2023) extend the AMR similarity-metric problem beyond a mere alignment-degree issue (i.e., how deeply two graph structures can be matched), and re-

conceptualize the metric as a three-module pipeline—pre-processing, alignment, and scoring—and introduce *Smatch++*, a new variant that systematically improves each of these components. Adopting *S²match* with fine-grained similarities, it additionally standardizes AMR graph inputs (including lower-casing, de-inverting edges, reification handling for non-core relations, and duplicate-triplet removal) before alignment, employs a more reliable alignment solver that guarantees optimal results, and computes final scores using finer-grained aspectual subgraph matching, such as location, named entities, negation, quantification, cause, and tense aspect structures. *S²match* thereby makes itself an extended and standardized version of the original *Smatch* framework.

Two more recent *Smatch*-based metrics have continued this line of research by improving conformity with AMR annotation guidelines and semantic theories. (Ki et al., 2024) introduce *Smatch#*, a metric built upon *Smatch++* that better captures the AMR notions of equivalence and non-equivalence through graph standardization. More recently, (Ki et al., 2026) extend *Smatch#* with *SEM-CAT*, further improving its conformance to truth-conditional semantics.

On a different line of research, (Groschwitz et al., 2023) introduce the Granular AMR Parsing Evaluation Suite (GrAPES), an evaluation benchmark for English AMR parsing. GrAPES was developed to enable a more comprehensive assessment of AMR parsers beyond a single aggregate score on a single test set. Specifically, it provides targeted test sets covering 36 linguistic phenomena, including pragmatic co-reference, ellipsis, PP (propositional phrase) attachment, rare words, unseen named entities, and structural generalization. Although GrAPES was originally proposed for evaluating AMR parsers, the idea of conducting fine-grained analyses of specific phenomena has since influenced the evaluation of AMR graph comparison methods.

(Shou and Lin, 2023), drawing inspiration from the embedding-based *WWLK*, use the transformer *BERT* to learn contextualized AMR graph representations through contrastive training and call it *AMRsim*. *AMRsim* is initialized with the uncased base version of *BERT*, and k-dimensional Graph Neural Networks (k-GNN adapter) are added on top of BERT layers to encode AMR structural information. Because human-annotated AMR graphs are costly to produce and limited in quantity, the model is instead trained on one million linearized silver graphs generated by the AMR parser *SPRING*. This architecture further improves correlation with human judgments and is robust to reification, synonymy, and variation in linearized graph sequences (Shou and Lin, 2023).

2 The Extended AMR Parsers and Metrics

Previous work has shown that the canonical AMR similarity metric *Smatch* outperforms string-based machine translation evaluation metrics such as *BLEU* within the WMT23 Metrics Shared Task framework. Building on this finding, the present study examines whether AMR-based metrics are broadly robust in evaluating the quality of system-generated translations. To this end, three additional AMR similarity metrics are selected, representing distinct methodological families: the BFS-based *SemBleu*, the alignment-based extension *Smatch++*, and the embedding-based *AMRsim*. *AMRsim* provides only segment-level graph-pair scores, so its system score is obtained by macro-averaging all segment scores. *SemBleu* additionally reports a document-level score, which can be treated as a micro system score. *Smatch++* outputs both segment-level scores and a micro-averaged system score. However, *Smatch++* returns F1 scores for different subgraphs based on the optimal alignment, as shown in Table 2. In this paper, the F1 scores across the different subgraphs are averaged to obtain a single F1 score for each segment and system. *SemBleu* uses the default setting with N=3 (unigram, bigram and trigram dependency sequences).

Using these AMR metrics, this study addresses several research questions. First, it examines whether the improved *Smatch++* aligns more closely with human judgments on translations than the original *Smatch*. Second, it investigates whether incorporating struc-

	metric	broad category	supervised	ref. free
baselines	BLEU	lexical overlap		
	F200SPBLEU	lexical overlap		
	CHRF	lexical overlap		
	BERTSCORE	embedding similarity		
	BLEURT	fine-tuned metric	✓	
	COMET	fine-tuned metric	✓	
	COMETKIWI	fine-tuned metric	✓	✓
	DOCWMT22COMETDA	fine-tuned metric	✓	
	DOCWMT22COMETKIWIDA	fine-tuned metric	✓	✓
	MS-COMET-QE-22	fine-tuned metric	✓	✓
	PRISMREF	MT-model metric	✓	
	PRISMSRC	MT-model metric	✓	✓
	RANDOM-SYSNAME	random baseline		✓
	YISI-1	embedding similarity		
primary submissions	CALIBRI-COMET22	fine-tuned metric	✓	
	CALIBRI-COMET22-QE	fine-tuned metric	✓	✓
	COMETOID22-WMT22	fine-tuned metric	✓	✓
	EBLEU	embedding similarity		
	EMBED_LLAMA	embedding similarity		
	GEMBA-MQM	LLM prompt-based metric		✓
	KG-BERTSCORE	embedding similarity		✓
	MATESE	fine-tuned metric	✓	
	MBR-METRICX-QE	fine-tuned metric	✓	✓
	MEE4	lexical & embedding similarity		
	METRICX-23	fine-tuned metric	✓	
	METRICX-23-QE	fine-tuned metric	✓	✓
	MRE-SCORE-LABSE-REGULAR	fine-tuned metric	✓	
	SESCORE	fine-tuned metric	✓	
	TOKENGRAM_F	lexical overlap		
	XCOMET-ENSEMBLE	fine-tuned metric	✓	
	XCOMET-QE-ENSEMBLE	fine-tuned metric	✓	✓
	XLSIM	fine-tuned metric	✓	

Figure 1: The 32 benchmark MT evaluation metrics of WMT 2023 Metrics Shared Task. Details for each MT metric please refer to (**?**).

tural dependencies, as in *SemBleu*, improves evaluation performance and how such a graph-based metric compares with the text-based translation quality metric *BLEU*. Third, it assess whether the embedding-based AMR metric *AMRsim* provides stronger translation evaluation performance than string-based AMR metrics and whether it can even outperform its text-based counterpart, *BERTScore* (Zhang et al., 2020). Most importantly, we examine whether improvements in AMR similarity metrics lead to more competitive AMR-based translation evaluation metrics overall.

AMR parser quality also plays a crucial role in the effectiveness of AMR-based similarity metrics for machine translation evaluation, since more accurate AMR graphs for system outputs and human references yield more reliable graph-pair similarity scores. In addition to the transition-based neural parser *Structured-BART Large* developed by IBM (Drozdov et al., 2022) (beam size = 1), this study also employs its substantially improved successor, *StructuredBART-MBSE* (Lee et al., 2022). The latter was further pre-trained on silver-standard data generated using the MBSE ensemble approach and decoded with a beam size of 10. It currently represents the best-performing AMR parser for which publicly available checkpoints are available.[1] *StructuredBART-MBSE* achieves a *Smatch* score of 84.4 on the AMR 3.0 test set, outperforming *Structured-BART Large* by 1.8 points. The motivation for using this stronger parsers is to maximize the reliability of AMR graph generation and thereby present the full capac-

[1]Both pre-trained models are available from the IBM GitHub repository: https://github.com/IBM/transition-amr-parser/tree/master. The specific checkpoints used in this study are *AMR3-structbart-L* and *AMR3-joint-ontowiki-seed43*.

Aspect	F1	Precision	Recall
AGENT	100.0	100.0	100.0
CAUSE	100.0	100.0	100.0
CONCEPT	85.71	85.71	85.71
FOCUS	0.0	0.0	0.0
INSTRUMENT	100.0	100.0	100.0
LOCATION	75.0	60.0	100.0
MATHS	100.0	100.0	100.0
NER	100.0	100.0	100.0
PATIENT	60.0	100.0	42.86
POLARITY	100.0	100.0	100.0
POSSESSION	100.0	100.0	100.0
PURPOSE	100.0	100.0	100.0
QUANTIFIER	100.0	100.0	100.0
QUESTION	100.0	100.0	100.0
SRL-core	60.0	100.0	42.86
TIME	100.0	100.0	100.0
TOPIC	100.0	100.0	100.0
WIKI	100.0	100.0	100.0
REENTRANCIES	100.0	100.0	100.0

Table 2: An example of *Smatch++*-generated results for the first segment of the GPT4-5shot machine-translated he→en data parsed by *StructuredBART-MBSE*.

ity of AMR-based similarity metrics in machine translation evaluation.

3 Results and Discussion

All results are computed following the same procedure as the previous study, using the WMT23 Metrics Shared Task datasets for Hebrew-to-English (he→en) and Chinese-to-English (zh→en), together with their corresponding MQM human quality ratings, correlation methods, and evaluation tools.

The resulting correlation scores of the different AMR similarity metrics are then compared with those of 32 established machine translation evaluation metrics evaluated on the same language pairs in WMT23.

3.1 AMR Metrics Scores Overview

The four AMR metrics produce substantially different ranges of graph similarity scores on the same datasets, as shown in Figure 2.

Smatch scores range from approximately 0.6 to 0.8 for he→en translations and from 0.3 to 0.6 for zh→en translations on average. Its extended variant, *Smatch++*, produces generally lower scores, as it evaluates the alignment of subgraphs rather than individual nodes and edges.

Task	Language	Granularity	Correlation	Weight
1	all	system	accuracy	2
2	he→en	system	Pearson	1
3	he→en	segment	Pearson	1
4	he→en	segment	acc-t	1
5	zh→en	system	Pearson	1
6	zh→en	segment	Pearson	1
7	zh→en	segment	acc-t	1

Table 3: Correlation tasks and weights used to compute the average correlation in this study, following the WMT23 Metrics Shared Task.

SemBleu produces the lowest graph similarity scores, consistent with the results in its original study, as it considers also longer trinode dependencies through BFS traversal. Its scores range from approximately 0.4 to 0.6 for he→en translations and from 0.3 to 0.45 for zh→en translations on average.

As a neural embedding-based approach, *AMRsim* produces the highest graph similarity scores, with both language pairs reaching approximately 0.8–0.9 on average. Unlike the other AMR similarity metrics, *AMRsim* also shows relatively little variation across machine translation systems, with the scores of different systems remaining close to one another when compared with the reference translations on graphs generated by both AMR parsers.

Scores from the other AMR metrics show varying degrees of variation across different machine translation systems, with *SemBleu* showing the largest variation, followed by *Smatch++*, which evaluates subgraphs that also incorporating some level of structural dependencies. Alignment-based *Smatch*, on the other hand, shows comparatively less variation. A notable pattern among the smatch variants is that they produce lower scores for zh→en graphs generated by *StructuredBART-MBSE* parser, particularly in the case of the basic *Smatch*. This may be related to the low quality of the zh→en reference translation (*refA*) (see the previous paper for details). Except for these two cases, graphs gen-

erated by *StructuredBART-MBSE* parser consistently achieve higher scores across metrics and language pairs. This suggests that *SemBleu* and *AMRsim* are less affected by the low quality of the zh→en reference translations than the *Smatch* variants.

The graph similarity scores produced by these AMR metrics reflect various characteristics of the metrics themselves. However, to determine whether these scores can effectively reflect the quality of the system translations, they need to be compared against the MQM human ratings of these translations. This correlation with human judgement is presented in the next section.

3.2 Correlation Ranking in WMT23

In the previous paper, the standard AMR metric *Smatch* achieved an average correlation (*avg-corr*) of 0.686 with MQM human ratings across the two language pairs and different tasks, using the human he→en reference translation B and zh→en reference translation A. This makes *Smatch* a structural-string-based MT evaluation metric that outperforms the textual-string-based MT metrics baselines *BLEU* and *CHRF* and text-only embedding-based MT metric *eBLEU*, as shown in Table 6. This study extends the previous work by evaluating three additional AMR similarity metrics: *Smatch++*, *SemBleu*, and *AMRsim*, as well as an additional, improved AMR parser pre-trained with silver-data generated using ensemble technique, *StructuredBART-MBSE*.

The performance of the four AMR metrics is first compared with established MT evaluation metrics in WMT23 using the original AMR parser, *StructuredBART Large*. The ranking results are shown in Table 5. As shown in table, the improved *Smatch* variant, *Smatch++*, performs almost identically to the original *Smatch*, achieving a global average correlation (*avg-corr*) that is only 0.001 higher. In contrast, *SemBleu* yields the lowest average correlation among the AMR metrics evaluated, performing only marginally better than the baseline *BLEU*. Among the four AMR similarity metrics, the neural metric *AMRsim* achieves the strongest correlation with human judgments, reaching a global average correlation of 0.698. This score is 0.011 higher than that of the second-best-performing metric, *Smatch++*. As a result, *AMRsim* outperforms embedding-based text-only MT evaluation metrics such as *MEE4* and *embed_llama*. However, it remains substantially below the text-based BERT embedding metric *BERTScore*, which achieves an *avg-corr* of 0.736 (0.038 higher) and ranks four positions higher.

Next, the four AMR metrics are applied to graphs from the same datasets, but generated by the superior AMR parser *StructuredBART-MBSE*. The correlation of the resulting scores with human ratings of MT quality is then compared with that of the established MT metrics in WMT23. The resulting ranking is shown in Table 4. As shown, using the better-parsed AMR graph representations not only increases the similarity scores of the four AMR metrics, but also improves their correlation with human ratings of translation quality on the same data.

SemBleu's average correlation increases from 0.681 to 0.688, raising its rank from 33rd to 30th and placing it above all text string-based MT metric baselines and text embedding-based *eBLEU*.

Smatch shows a smaller improvement, likely due to its sensitivity to the low-quality reference translation in the zh→en data. Its average correlation increases from 0.686 to 0.691, placing it above one more text-based embedding MT metric, *MEE4*.

Smatch++, however, benefits more substantially from the better-parsed graphs. Its improved performance suggests that its standardization preprocessing and subgraph-based scoring help to capture more semantic similarities. It rises to 25th place, three ranks above *Smatch*, as its average correlation increases from 0.687 to 0.699 (a gain of 0.012), which brings it level with *AMRsim* using the original AMR parser. This makes it a better MT evaluation metric than all text string-based ones.

AMRsim remains the best-performing AMR-based metric, achieving an average correlation with human MT ratings of 0.709, an improvement of 0.011 over the results obtained with the previous parser. Despite this notable increase, its ranking improves by only one position, rising from 24th to 23rd place, as performance differences between metrics



Figure 2: System level scores for all AMR metrics across language pairs and AMR parsers. 'Large' refers to *Structured-BART Large* parser and 'ensemble' refers to *StructuredBART-MBSE* parser.

become more pronounced at the higher end of the ranking. One possible explanation is that *AMRsim* already assigns similarity scores around 0.9 to AMR graphs generated by the lower-performing parser, leaving limited room for additional gains when using outputs from the stronger parser, as is observed in Figure 2.

3.3 Discussion

This study investigates the effectiveness of AMR-based metrics for evaluating translation quality. It uses three additional AMR similarity metrics alongside the basic *Smatch*, together with an extra stronger AMR parser, *StructuredBART-MBSE*. The results show that two improvements further raise the performance of AMR-based metrics in the WMT23 MT evaluation from 28th to 23rd place: more accurate graphs produced by a stronger parser, and an AMR similarity metric that better captures the semantic similarity between graphs. However, these metrics still remain distant from the embedding-based MT metrics baseline *BERTscore*, and the gap to the top fine-tuning-based MT evaluation metrics is even larger.

The best-performing AMR graph-based MT metric *AMRsim* aligns better with human ratings than all text-based MT metrics that rely on lexical overlap. It also outperforms several embedding-based MT metrics, such as *eBLEU*, *MEE4*, and *embed_llama*, and even the fine-tuned metric *XLsim*. However, *AMRsim* still falls far behind its text-only counterpart *BERTscore*, even though both are built on BERT.

This does not neccessarily mean that graphs are less useful than text for MT evaluation. The string-based AMR metrics *Smatch*, *Smatch++*, and *SemBleu*, in contrast, all align better with human ratings than their corresponding string-based MT metrics that rely on only text, and even the weakest of them, *SemBleu*, outperforms both *BLEU* and its embedding-based variant *eBLEU*. This inconsistency suggests that the weaker performance of *AMRsim* is unlikely to stem from the AMR graphs themselves, since the string-based AMR metrics use the same kind of parsed graphs and benefit from them. A more plausible explanation lies in how *AMRsim* is trained, for example parsing errors in the silver training graphs, or a tendency of the model to focus on the similarities between graphs rather than their differences, which would also explain its high average scores. Indeed, the average similarity scores that *AMRsim* assigns to graph pairs remain around 0.9 across MT systems, language pairs, and parsers, which suggests that it compresses differences between pairs and therefore struggles to separate good translations from poor ones.

Another finding is that *SemBleu* performs worse than the two *Smatch* variants, so incorporating structural dependencies through BFS traversal appears to hurt rather than help MT evaluation. This may indicate that comparing semantic representations more strictly, by including structural dependencies, makes a metric less robust to structural variation between sentences with similar meanings.

On the other hand, *Smatch++* improves more substantially than basic *Smatch* when the *StructuredBART-MBSE* parser is used, which points to the value of standardizing the graphs before the alignment search, using a search that is guaranteed to find the optimal alignment, and scoring with subgraph mappings: together, they do help reveal the semantic similarities between sentence pairs and lead to better agreement with human judgments.

4 Conclusion

Overall, AMR-based metrics are not yet competitive with the strongest MT evaluation metrics: even with a stronger parser and a better graph similarity metric, they trail *BERTscore* and, by a wider margin, the top fine-tuned MT metrics.

Nevertheless, the results are encouraging. The string-based AMR metrics already outperform comparable text-based ones, and the improvements from *StructuredBART-MBSE* and *Smatch++* show that better graphs and matching procedures translate directly into better alignment with human judgments on translation qualities. The AMR parser still has considerable room for improvement, and so do embedding-based AMR similarity measures.The weakness of *AMRsim* seems to lie in how graph information is encoded. Incorporating this information more effectively into pre-trained transformers is a promising direc-

lang: level: corr_fcn: metric	avg-corr		he-en,zh-en sys accuracy task1		he-en sys pearson task2		he-en seg pearson task3		he-en seg acc-t task4		zh-en sys pearson task5		zh-en seg pearson task6		zh-en seg acc-t task7	
XCOMET-Ensemble	**1**	**0.815**	3	0.923	4	0.950	**1**	**0.556**	**1**	**0.586**	8	0.927	**1**	**0.650**	**1**	**0.543**
MetricX-23	2	0.800	4	0.907	13	0.910	2	0.548	2	0.577	16	0.873	4	0.625	4	0.531
XCOMET-QE-Ensemble*	3	0.793	6	0.891	14	0.909	4	0.498	5	0.554	13	0.892	3	0.647	3	0.533
GEMBA-MQM*	4	0.790	**1**	**0.923**	6	0.939	7	0.401	4	0.564	**1**	**0.991**	9	0.449	7	0.522
MetricX-23-QE*	5	0.787	8	0.880	24	0.858	3	0.520	3	0.564	18	0.859	2	0.647	5	0.527
MaTESe	6	0.782	2	0.923	16	0.906	5	0.459	7	0.550	14	0.889	6	0.511	24	0.479
mbr-metricx-qe*	7	0.773	19	0.858	11	0.915	6	0.411	6	0.553	6	0.936	7	0.489	2	0.537
CometKiwi*	8	0.768	10	0.874	23	0.860	12	0.387	8	0.544	3	0.963	11	0.442	6	0.525
KG-BERTScore*	9	0.765	16	0.863	15	0.908	14	0.382	9	0.537	4	0.962	12	0.430	9	0.516
COMET	10	0.764	9	0.874	5	0.940	8	0.401	10	0.532	12	0.898	13	0.396	11	0.514
cometoid22-wmt22*	11	0.760	14	0.869	27	0.839	17	0.365	15	0.515	5	0.940	8	0.479	10	0.515
Calibri-COMET22	12	0.759	5	0.896	8	0.930	9	0.401	16	0.515	17	0.863	14	0.396	27	0.474
BLEURT-20	13	0.757	15	0.863	7	0.937	15	0.382	14	0.519	15	0.880	16	0.378	8	0.518
sescoreX	14	0.753	17	0.863	18	0.901	13	0.385	23	0.484	23	0.797	5	0.536	14	0.499
docWMT22CometKiwiDA*	15	0.750	12	0.874	17	0.906	22	0.286	22	0.489	2	0.965	15	0.387	17	0.493
docWMT22CometDA	16	0.750	11	0.874	9	0.922	18	0.339	20	0.497	9	0.907	18	0.353	18	0.493
mre-score-labse-regular	17	0.746	7	0.885	**1**	**0.958**	16	0.378	13	0.522	10	0.903	27	0.145	22	0.481
YiSi-1	18	0.745	18	0.858	10	0.917	10	0.395	11	0.529	20	0.823	20	0.290	12	0.504
Calibri-COMET22-QE*	19	0.743	23	0.836	31	0.778	11	0.395	18	0.506	7	0.934	10	0.443	19	0.491
BERTscore	20	0.736	13	0.869	19	0.895	19	0.335	17	0.515	21	0.810	22	0.236	15	0.499
MS-COMET-QE-22*	21	0.729	21	0.847	34	0.721	21	0.295	19	0.498	11	0.901	17	0.367	16	0.498
prismRef	22	0.727	25	0.836	2	0.956	20	0.319	12	0.528	29	0.762	24	0.183	13	0.504
AMR_sim	23	0.709	26	0.831	3	0.951	30	0.223	21	0.495	19	0.826	32	0.096	30	0.465
XLsim	24	0.705	22	0.842	20	0.887	27	0.233	24	0.480	24	0.796	29	0.111	31	0.464
AMR_sm++	25	0.699	30	0.787	28	0.836	23	0.281	26	0.468	32	0.739	19	0.337	32	0.456
embed_llama	26	0.694	24	0.836	26	0.841	33	0.215	35	0.430	27	0.785	26	0.161	34	0.447
tokengram_F	27	0.692	28	0.803	22	0.878	29	0.226	27	0.461	25	0.795	34	0.060	21	0.485
AMR_sm	28	0.691	34	0.776	25	0.845	25	0.255	25	0.476	30	0.743	21	0.255	33	0.455
MEE4	29	0.691	27	0.809	21	0.879	24	0.256	34	0.441	31	0.743	31	0.105	23	0.480
AMR_bleu	30	0.688	33	0.776	29	0.829	26	0.240	28	0.460	26	0.791	25	0.176	26	0.475
f200spBLEU	31	0.685	29	0.798	30	0.805	28	0.230	30	0.447	28	0.772	30	0.108	25	0.476
eBLEU	32	0.683	20	0.852	12	0.911	35	0.131	31	0.445	34	0.727	36	-0.084	28	0.473
chrF	33	0.682	32	0.787	32	0.776	31	0.221	29	0.460	22	0.809	33	0.063	20	0.485
BLEU	34	0.676	31	0.787	33	0.769	32	0.220	32	0.442	33	0.734	28	0.119	29	0.472
Random-sysname*	35	0.520	35	0.585	35	0.209	36	0.041	36	0.428	35	0.093	35	0.018	36	0.381
prismSrc*	36	0.462	36	0.432	36	-0.017	34	0.140	33	0.441	36	-0.406	23	0.223	35	0.421

Table 4: WMT23 evaluation metrics ranking with four AMR metrics included, using graphs parsed by the AMR parser *StructuredBART-MBSE*. AMR *Smatch* is denoted as *AMR_sm*, *Smatch++* as *AMR_sm++*, *SemBleu* as *AMR_bleu*, and *AMRsim* as *AMR_sim*. MT metrics marked with * are reference-free; underlined are baselines.

tion.

Future work includes extending the investigation of the viability of AMR-based metrics as MT evaluation metrics by using more language pairs. AMR-based metrics also have the potential to support reference-free MT evaluation once powerful multilingual AMR parsers become available.

References

Rafael T. Anchieta, Marco A. S. Cabezudo, and Thiago A. S. Pardo. 2019. Sema: an extended semantic evaluation metric for amr.

Xuefeng Bai, Yulong Chen, and Yue Zhang. 2022. Graph pre-training for AMR parsing and generation. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6001–6015, Dublin, Ireland. Association for Computational Linguistics.

Guntis Barzdins and Didzis Gosko. 2016. RIGA at SemEval-2016 task 8: Impact of Smatch extensions and character-level neural translation on AMR parsing accuracy. In *Proceedings of the 10th International Workshop on Semantic Evaluation (SemEval-2016)*, pages 1143–1147, San Diego, California. Association for Computational Linguistics.

Petr Baudiš, Jan Pichl, Tomáš Vyskočil, and Jan Šedivý. 2016. Sentence pair scoring: Towards unified framework for text comprehension.

Michele Bevilacqua, Rexhina Blloshmi, and Roberto Navigli. 2021. One SPRING to rule them both: Symmetric AMR semantic parsing and generation without a complex pipeline. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 12564–12573.

Maja Buljan, Stephan Oepen, and Lilja Øvrelid. 2026. Context is (almost) everything: Llama-3 on structured output and amr parsing. In *Proceedings of the Fifteenth Language Resources and Evaluation Conference (LREC 2026)*, volume 11, pages 11679–11698.

Deng Cai and Wai Lam. 2020. AMR parsing via graph-sequence iterative inference. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 1290–1301, Online. Association for Computational Linguistics.

Shu Cai and Kevin Knight. 2013. Smatch: an evaluation metric for semantic feature structures. In *Proceedings of the 51st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 748–752, Sofia, Bulgaria. Association for Computational Linguistics.

Hyung Won Chung, Le Hou, Shayne Longpre, Barret Zoph, Yi Tay, William Fedus, Yunxuan Li, Xuezhi Wang, Mostafa Dehghani, Siddhartha Brahma, Albert Webson, Shixiang Shane Gu, Zhuyun Dai, Mirac Suzgun, Xinyun Chen, Aakanksha Chowdhery, Alex Castro-Ros, Marie Pellat, Kevin Robinson, Dasha Valter, Sharan Narang, Gaurav Mishra, Adams Yu, Vincent Zhao, Yanping Huang, Andrew Dai, Hongkun Yu, Slav Petrov, Ed H. Chi, Jeff Dean, Jacob Devlin, Adam Roberts, Denny Zhou, Quoc V. Le, and Jason Wei. 2022. Scaling instruction-finetuned language models.

Marco Damonte, Shay B. Cohen, and Giorgio Satta. 2017. An incremental parser for Abstract Meaning Representation. In *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 1, Long Papers*, pages 536–546, Valencia, Spain. Association for Computational Linguistics.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding.

Andrew Drozdov, Jiawei Zhou, Radu Florian, Andrew McCallum, Tahira Naseem, Yoon Kim, and Ramon Fernandez Astudillo. 2022. Inducing and using alignments for transition-based amr parsing.

Jeffrey Flanigan, Sam Thomson, Jaime Carbonell, Chris Dyer, and Noah A. Smith. 2014. A discriminative graph-based parser for the Abstract Meaning Representation. In *Proceedings of the 52nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1426–1436, Baltimore, Maryland. Association for Computational Linguistics.

Markus Freitag, Nitika Mathur, Chi-kiu Lo, Eleftherios Avramidis, Ricardo Rei, Brian Thompson, Tom Kocmi, Frederic Blain, Daniel Deutsch, Craig Stewart, Chrysoula Zerva, Sheila Castilho, Alon Lavie, and George Foster. 2023. Results of WMT23 metrics shared task: Metrics might be guilty but references are not innocent. In *Proceedings of the Eighth Conference on Machine Translation*, pages 578–628, Singapore. Association for Computational Linguistics.

Jonas Groschwitz, Shay B. Cohen, Lucia Donatelli, and Meaghan Fowlie. 2023. Amr parsing is far from solved: Grapes, the granular amr parsing evaluation suite.

Hardy and Andreas Vlachos. 2018. Guided neural language generation for abstractive summarization using Abstract Meaning Representation. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 768–773, Brussels, Belgium. Association for Computational Linguistics.

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XCOMET-Ensemble	**1**	**0.815**	3	0.923	4	0.950	**1**	**0.556**	**1**	**0.586**	8	0.927	**1**	**0.650**	**1**	**0.543**
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XCOMET-Ensemble	**1**	**0.815**	**1**	**0.923**	**1**	**0.950**	**1**	**0.556**	**1**	**0.586**	2	0.927	**1**	**0.650**	**1**	**0.543**
MetricX-23	2	0.800	**1**	**0.907**	2	0.910	**1**	**0.548**	2	0.577	4	0.873	2	0.625	3	0.531
XCOMET-QE-Ensemble*	2	0.793	2	0.891	2	0.909	3	0.498	4	0.554	3	0.892	**1**	**0.647**	3	0.533
GEMBA-MQM*	2	0.790	**1**	**0.923**	2	0.939	5	0.401	3	0.564	**1**	**0.991**	6	0.449	5	0.522
MetricX-23-QE*	2	0.787	2	0.880	3	0.858	2	0.520	3	0.564	4	0.859	**1**	**0.647**	4	0.527
MaTESe	3	0.782	**1**	**0.923**	2	0.906	4	0.459	5	0.550	3	0.889	4	0.511	12	0.479
mbr-metricx-qe*	3	0.773	2	0.858	2	0.915	5	0.411	4	0.553	2	0.936	5	0.489	2	0.537
CometKiwi*	3	0.768	2	0.874	3	0.860	7	0.387	6	0.544	2	0.963	6	0.442	4	0.525
KG-BERTScore*	4	0.765	2	0.863	2	0.908	8	0.382	7	0.537	2	0.962	7	0.430	6	0.516
COMET	4	0.764	2	0.874	2	0.940	6	0.401	8	0.532	3	0.898	8	0.396	7	0.514
cometoid22-wmt22*	4	0.760	2	0.869	4	0.839	9	0.365	11	0.515	2	0.940	5	0.479	7	0.515
Calibri-COMET22	4	0.759	**1**	**0.896**	2	0.930	6	0.401	11	0.515	4	0.863	8	0.396	14	0.474
BLEURT-20	5	0.757	2	0.863	2	0.937	8	0.382	10	0.519	4	0.880	9	0.378	6	0.518
sescoreX	5	0.753	2	0.863	3	0.901	7	0.385	15	0.484	5	0.797	3	0.536	9	0.499
docWMT22CometKiwiDA*	5	0.750	2	0.874	2	0.906	12	0.286	14	0.489	2	0.965	8	0.387	10	0.493
docWMT22CometDA	5	0.750	2	0.874	2	0.922	10	0.339	13	0.497	3	0.907	10	0.353	10	0.493
mre-score-labse-regular	5	0.746	2	0.885	**1**	**0.958**	8	0.378	10	0.522	3	0.903	16	0.145	12	0.481
YiSi-1	5	0.745	2	0.858	2	0.917	6	0.395	8	0.529	4	0.823	11	0.290	8	0.504
Calibri-COMET22-QE*	5	0.743	2	0.836	4	0.778	6	0.395	12	0.506	2	0.934	6	0.443	10	0.491
BERTscore	6	0.736	2	0.869	3	0.895	10	0.335	11	0.515	5	0.810	12	0.236	9	0.499
MS-COMET-QE-22*	6	0.729	2	0.847	5	0.721	12	0.295	13	0.498	3	0.901	9	0.367	9	0.498
prismRef	6	0.727	2	0.836	**1**	**0.956**	11	0.319	9	0.528	6	0.762	14	0.183	8	0.504
XLsim	7	0.705	2	0.842	3	0.887	14	0.233	16	0.480	5	0.796	17	0.111	16	0.464
embed_llama	7	0.694	2	0.836	4	0.841	16	0.215	21	0.430	5	0.785	15	0.161	17	0.447
tokengram_F	7	0.692	3	0.803	3	0.878	15	0.226	17	0.461	5	0.795	20	0.060	11	0.485
MEE4	8	0.691	3	0.809	3	0.879	13	0.256	20	0.441	6	0.743	18	0.105	12	0.480
AMR_sm	8	0.686	3	0.803	3	0.846	14	0.245	18	0.454	8	0.605	13	0.217	15	0.468
f200spBLEU	8	0.685	3	0.798	4	0.805	15	0.230	19	0.447	6	0.772	18	0.108	13	0.476
eBLEU	8	0.683	2	0.852	2	0.911	17	0.131	19	0.445	7	0.727	22	-0.084	14	0.473
chrF	8	0.682	3	0.787	4	0.776	16	0.221	17	0.460	5	0.809	19	0.063	11	0.485
BLEU	8	0.676	3	0.787	5	0.769	16	0.220	20	0.442	7	0.734	17	0.119	14	0.472
Random-sysname*	9	0.520	4	0.585	6	0.209	18	0.041	21	0.428	8	0.093	21	0.018	19	0.381
prismSrc*	9	0.462	4	0.432	6	-0.017	17	0.140	20	0.441	9	-0.406	13	0.223	18	0.421

Table 6: WMT23 evaluation metrics ranking with AMR Smatch scores included. AMR Smatch is written as *AMR_sm*. MT metrics marked with * are reference-free.

Shu Han Ho. 2025. Evaluation of finetuned llms in amr parsing.

Yilun Hua, Zhaoyuan Deng, and Zhijie Xu. 2022. AMRTVSumm: AMR-augmented hierarchical network for TV transcript summarization. In *Proceedings of the Workshop on Automatic Summarization for Creative Writing*, pages 36–43, Gyeongju, Republic of Korea. Association for Computational Linguistics.

Fuad Issa, Marco Damonte, Shay B. Cohen, Xiaohui Yan, and Yi Chang. 2018. Abstract Meaning Representation for paraphrase detection. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, pages 442–452, New Orleans, Louisiana. Association for Computational Linguistics.

Pavan Kapanipathi, Ibrahim Abdelaziz, Srinivas Ravishankar, Salim Roukos, Alexander Gray, Ramón Fernandez Astudillo, Maria Chang, Cristina Cornelio, Saswati Dana, Achille Fokoue, Dinesh Garg, Alfio Gliozzo, Sairam Gurajada, Hima Karanam, Naweed Khan, Dinesh Khandelwal, Young-Suk Lee, Yunyao Li, Francois Luus, Ndivhuwo Makondo, Nandana Mihindukulasooriya, Tahira Naseem, Sumit Neelam, Lucian Popa, Revanth Gangi Reddy, Ryan Riegel, Gaetano Rossiello, Udit Sharma, G P Shrivatsa Bhargav, and Mo Yu. 2021. Leveraging Abstract Meaning Representation for knowledge base question answering. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pages 3884–3894, Online. Association for Computational Linguistics.

Kyung Seo Ki, Bugeun Kim, and Gahgene Gweon. 2024. Inspecting soundness of AMR similarity metrics in terms of equivalence and inequivalence. In *Proceedings of the 13th Joint Conference on Lexical and Computational Semantics (*SEM 2024)*, pages 402–409, Mexico City, Mexico. Association for Computational Linguistics.

Kyung Seo Ki, Bugeun Kim, and Gahgene Gweon. 2026. Semantic evaluation metric conforming to amr theory (semcat): a new similarity metric for abstract meaning representation. *Lang. Resour. Eval.*, 60(2).

Ioannis Konstas, Srinivasan Iyer, Mark Yatskar, Yejin Choi, and Luke Zettlemoyer. 2017. Neural AMR: Sequence-to-sequence models for parsing and generation. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 146–157, Vancouver, Canada. Association for Computational Linguistics.

Young-Suk Lee, Ramón Astudillo, Hoang Thanh Lam, Tahira Naseem, Radu Florian, and Salim Roukos. 2022. Maximum Bayes Smatch ensemble distillation for AMR parsing. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 5379–5392, Seattle, United States. Association for Computational Linguistics.

Young-Suk Lee, Ramón Fernandez Astudillo, Radu Florian, Tahira Naseem, and Salim Roukos. 2023. Amr parsing with instruction fine-tuned pre-trained language models.

Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Ves Stoyanov, and Luke Zettlemoyer. 2019. Bart: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension.

Changmao Li and Jeffrey Flanigan. 2022. Improving neural machine translation with the Abstract Meaning Representation by combining graph and sequence transformers. In *Proceedings of the 2nd Workshop on Deep Learning on Graphs for Natural Language Processing (DLG4NLP 2022)*, pages 12–21, Seattle, Washington. Association for Computational Linguistics.

Yanming Li and Meaghan Fowlie. 2025. Gpt makes a poor amr parser. *Journal for Language Technology and Computational Linguistics*, 38(2):43–76.

Song Linfeng, Gildea Daniel, Zhang Yue, Wang Zhiguo, and Su Jinsong. 2019. Semantic neural machine translation using amr. *Transactions of the Association for Computational Linguistics*, 7:19–31.

Marco Marelli, Stefano Menini, Marco Baroni, Luisa Bentivogli, Raffaella Bernardi, and Roberto Zamparelli. 2014. A SICK cure for the evaluation of compositional distributional semantic models. In *Proceedings of the Ninth International Conference on Language Resources and Evaluation (LREC'14)*, pages 216–223, Reykjavik, Iceland. European Language Resources Association (ELRA).

Long HB Nguyen, Viet H Pham, and Dien Dinh. 2021. Improving neural machine translation with amr semantic graphs. *Mathematical Problems in Engineering*, 2021(1):9939389.

Juri Opitz. 2023. Smatch++: Standardized and extended evaluation of semantic graphs. In *Findings of the Association for Computational Linguistics: EACL 2023*, page 1595–1607. Association for Computational Linguistics.

Juri Opitz, Angel Daza, and Anette Frank. 2021. Weisfeiler-leman in the bamboo: Novel AMR graph metrics and a benchmark for AMR graph

similarity. *Transactions of the Association for Computational Linguistics*, 9:1425–1441.

Juri Opitz, Letitia Parcalabescu, and Anette Frank. 2020. Amr similarity metrics from principles. *Transactions of the Association for Computational Linguistics*, 8:522–538.

Hyeonchu Park, Byungjun Kim, and Bugeun Kim. 2025. DART: An AIGT detector using AMR of rephrased text. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 2: Short Papers)*, pages 710–721, Albuquerque, New Mexico. Association for Computational Linguistics.

Xiaochang Peng, Chuan Wang, Daniel Gildea, and Nianwen Xue. 2017. Addressing the data sparsity issue in neural AMR parsing. In *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 1, Long Papers*, pages 366–375, Valencia, Spain. Association for Computational Linguistics.

Jeffrey Pennington, Richard Socher, and Christopher Manning. 2014. GloVe: Global vectors for word representation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1532–1543, Doha, Qatar. Association for Computational Linguistics.

Sudha Rao, Daniel Marcu, Kevin Knight, and Hal Daumé III. 2017. Biomedical event extraction using Abstract Meaning Representation. In *Proceedings of the 16th BioNLP Workshop*, pages 126–135, Vancouver, Canada,. Association for Computational Linguistics.

Ziyi Shou and Fangzhen Lin. 2023. Evaluate AMR graph similarity via self-supervised learning. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 16112–16123, Toronto, Canada. Association for Computational Linguistics.

Linfeng Song and Daniel Gildea. 2019. SemBleu: A robust metric for AMR parsing evaluation. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4547–4552, Florence, Italy. Association for Computational Linguistics.

Afonso Sousa and Henrique Lopes Cardoso. 2025. Sapg: Semantically-aware paraphrase generation with amr graphs. In *ICAART (2)*, pages 861–871.

Matteo Togninalli, Elisabetta Ghisu, Felipe Llinares-López, Bastian Rieck, and Karsten Borgwardt. 2019. Wasserstein weisfeiler-lehman graph kernels.

Pavlo Vasylenko, Pere-Lluís Huguet Cabot, Abelardo Carlos Martínez Lorenzo, and Roberto Navigli. 2023. Incorporating graph information in transformer-based amr parsing.

Chuan Wang, Sameer Pradhan, Xiaoman Pan, Heng Ji, and Nianwen Xue. 2016. CAMR at SemEval-2016 task 8: An extended transition-based AMR parser. In *Proceedings of the 10th International Workshop on Semantic Evaluation (SemEval-2016)*, pages 1173–1178, San Diego, California. Association for Computational Linguistics.

Shira Wein and Nathan Schneider. 2022. Accounting for language effect in the evaluation of cross-lingual AMR parsers. In *Proceedings of the 29th International Conference on Computational Linguistics*, pages 3824–3834, Gyeongju, Republic of Korea. International Committee on Computational Linguistics.

Sheng Zhang, Xutai Ma, Kevin Duh, and Benjamin Van Durme. 2019. AMR parsing as sequence-to-graph transduction. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 80–94, Florence, Italy. Association for Computational Linguistics.

Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. 2020. Bertscore: Evaluating text generation with bert.

Jiawei Zhou, Tahira Naseem, Ramón Fernandez Astudillo, and Radu Florian. 2021a. AMR parsing with action-pointer transformer. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 5585–5598, Online. Association for Computational Linguistics.

Jiawei Zhou, Tahira Naseem, Ramón Fernandez Astudillo, Young-Suk Lee, Radu Florian, and Salim Roukos. 2021b. Structure-aware fine-tuning of sequence-to-sequence transformers for transition-based AMR parsing. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 6279–6290, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.

Weiran Zhou and Guizhi Zhang. 2026. Cross-language english neural machine translation based on amr and graph cyclic neural network. *Journal of Cases on Information Technology (JCIT)*, 28(1):1–17.